

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1430

Stuttgart Ricuskey 100 kV
February 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1430. The Interconnection Customer proposes a 25.3 MW battery energy storage facility interconnecting at the Stuttgart Ricuskey 100 kV substation in the Illinois area. The request was received on March 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 561 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$756,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FRENCH LANDING DAM – HOUMA 100kV	MOSBY – MOUNT POLLEY 100kV	132.1	122	8.9
CARBON CTR 500/345 kV Autotransformer	BLACKROCK – SHAWVILLE STA 345kV	126.7	1,998	6.2
SHAWVILLE STA – CARBON CTR 500kV	BLACKROCK – SHAWVILLE STATION SUBSTATION 345kV	119.4	2,111	6.2
MOUNT POLLEY – PAMPANOS 100kV line	TIDELANDS SUBSTATION 230/1kV Auto	115.1	149	17.3
ORCHARD ROAD – SHAWVILLE STA 500kV	SHAWVILLE STA – SCOTT 345kV	114.6	2,944	7.8
PINEVILLE DELIVERY 1 SUBSTATION – KIMBERLY 100kV Ckt 1	PINEVILLE DELIVERY 1 – KIMBERLY 100kV Line	109.1	143	5.7
KIMBERLY – PINEVILLE DELIVERY 1 100kV Cir. 2	PINEVILLE DELIVERY 1 – KIMBERLY 100kV Circuit 1	109.1	143	5.7
SHAWVILLE STATION – BLACKROCK SUBSTATION 345kV Line	CARBON CTR 500 kV/345 kV TX	104.7	2,489	7.1
BLACKROCK – CALDWELL 345kV	CARBON CENTER 500/345kV Transformer	103.3	1,935	5.3
CORONET – FT HUMBUG 500kV	ACADIA – MONTOUR STA 500kV	103.2	3,346	8.2
MADISON PARK – HENRY D KING 100kV	CARRIZO PLAINS – SPILLS 100kV line	100.7	128	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FRENCH LANDING DAM – HOUMA 100kV	Rebuild line with higher-capacity conductor	\$1,668,000	4% (\$68,000)
CARBON CTR 500/345 kV XFMR	Replace transformer with higher-rated unit	\$21,031,000	0% (\$27,000)
SHAWVILLE STA – CARBON CTR 500kV	Rebuild line with higher-capacity conductor	\$138,815,000	0% (\$177,000)
PAMPANOS – MOUNT POLLEY 100kV	Rebuild line with higher-capacity conductor	\$919,000	7% (\$62,000)
ORCHARD ROAD – SHAWVILLE STA 500kV	Rebuild line with higher-capacity conductor	\$184,425,000	0% (\$236,000)
PINEVILLE DELIVERY 1 – KIMBERLY 100kV	Reconductor line	\$407,000	2% (\$9,000)
KIMBERLY – PINEVILLE DELIVERY 1 100kV Circuit 2	Reconductor line	\$616,000	2% (\$14,000)
SHAWVILLE STATION – BLACKROCK SUBSTATION 345kV Line	Reconductor line	\$12,924,000	0% (\$17,000)
BLACKROCK – CALDWELL 345kV	Reconductor line	\$14,716,000	0% (\$19,000)
CORONET – FORT HUBUG 500kV	Reconductor line	\$79,453,000	0% (\$101,000)
HENRY D KING – MADISON PARK 100kV	Reconductor line	\$279,000	9% (\$26,000)
	Total allocated to project		\$756,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.